Patricio F. Mendez

Department of Chemical and Materials Engineering
University of Alberta
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Education

Massachusetts Institute of Technology (MIT) Cambridge, MA

Doctor of Philosophy (Ph.D.) Materials Engineering 1999

Order of Magnitude Scaling of Complex Engineering Problems and its Application to High Productivity Arc Welding. Advisor: Prof. T. W. Eagar.

Established foundations for scaling problems with many coupled physical phenomena. Identified the root cause of welding speed limit in arc welding and developed a quantitative criterion. Coursework GPA 4.9/5. Recipient of Rocca Fellowship from MIT. Thesis work received the Easterling Award from the International Institute of Welding and the Jennings Award from the American Welding Society.

Minor Management of Technology, Sloan School of Management 1994-1995

Courses on Management of Technology, Product Design, and Negotiation.

Master of Science (M.S.) Materials Science and Engineering 1995

Joining Metals Using Semi-Solid Slurries

Invented and demonstrated a new method for joining metals. Thesis work received a US patent and was foundational to the startup Semi-Solid Technologies.

University of Buenos Aires (UBA) Buenos Aires, Argentina

Mechanical Engineer 1992

Major in Machine Design. Six-year program with a foundation of 6 semesters of calculus, 4 semesters of linear algebra, 5 semesters of physics, 6 semesters of solid mechanics, 5 semesters of electrical engineering, and two semesters each of fluid mechanics, heat transfer, and dynamics. In addition of multiple classes on materials, manufacturing, and machine design, the program also included economics and law. Highest ranked graduating student (equivalent to Summa Cum Laude).

Professional Experience

University of Alberta (UofA) Edmonton, AB

Professor, Department of Chemical and Materials Engineering 2013 – Present

Weldco/Industry Chair in Welding and Joining 2009 – Present

Director, Canadian Centre for Welding and Joining (CCWJ) 2009 – Present

Associate Professor, Department of Chemical and Materials Engineering 2009 – 2013

Created the CCWJ, one of six official research centers at Faculty of Engineering. The CCWJ is currently a $5.4M facility involving multiple faculty, students, and topics associated with welding and joining, with a focus on physics, metallurgy, and automation, and support from industry and federal and provincial governments. Since its inception, the CCWJ has raised $10.3M for research, graduated 54 graduate students, involved in research 279 visiting, undergraduate, or high-school students involving 28 nationalities, supported two startups, organized 69 seminars for 1681 members of industry, received multiple international awards, and its graduates participate in positions of leadership in the industry and professional societies.

Colorado School of Mines (CSM) Golden, CO
Affiliate Faculty, Metallurgical and Materials Engineering 2019 – present
Co-advising PhD student on hydrogen effects on elastic modulus of stainless steel.

Assistant Professor, Metallurgical and Materials Engineering 2004 – 2008
Responsibilities included teaching and research associated with welding and casting. Appointed "Key Professor" by the Foundry Education Foundation (FEF) of the US. Commissioned second largest university foundry in the US involving 4 induction furnaces (up to 500 lb for ferrous, 150 lb for aluminum), two gas furnaces, an inert gas arc furnace and capabilities for sand casting, sand and clay testing, mold making, additive manufacturing patterns, wax patterns, investment casting, and jewelry casting. Hosted open Friday casting activities open to community. Foundry class involving solidification theory and hands-on casting was one of two most popular classes in the department. The foundry program produced approximately 30 engineers per year with first-hand understanding of casting, more than any other university in North America. Raised $1.1M in external research funds, including NSF CAREER Award. Graduated 4 Ph.D. and 3 M.S. with focus on welding and casting.

PGS Technical Consulting Edmonton, AB
Owner 2014– present
Technical consulting activities. Focus on knowledge transfer, implementing existing state of the art-research into commercial products and processes.

Exponent Natick, MA
Senior Engineer, Mechanics and Materials Practice 2004
Engineer, Mechanics and Materials Practice 2002 – 2004
Engineering consultant in process simulation, materials issues, and biomedical devices, with a focus on coronary stents. Generated $570K in consulting income involving more than 70 cases.

Semi-Solid Technologies, Inc. (SST) Cambridge, MA
President 2000 – 2006
Vice-President, Co-founder, and Co-owner 1995 – 2000
Invented and commercialized technologies based on semi-solid metal processing, receiving 6 US patents and 3 international patents. Developed, demonstrated, and patented one of the first additive manufacturing technologies for metals. Invented and commercialized semi-solid aluminum die-casting process used to make fuel rails under license to Gibbs Die Casting in Kentucky. All Ford Focus Zetec engines around 1998-2002 were featured with this component. In addition to licensing revenues, raised $1M from government and industrial sources for development of casting and additive manufacturing processes.

Massachusetts Institute of Technology (MIT) Cambridge, MA
Research Affiliate, Department of Materials Science and Engineering 2002 – 2004
Expanded research on scaling modeling and welding trends in aeronautics. Co-advised M.S. student in the Department of Materials Science and Engineering (With Prof. Eagar and Prof. Masubuchi).

Postdoctoral Associate, Department of Materials Science and Engineering 1999 – 2002
Investigated statistics-based scaling modeling techniques and welding trends in aeronautics. Co-developed code SLAW to generate physics-based scaling laws from statistical data. Focused on arc plasmas, weld pool fluid/thermal interactions, and metal transfer in GMAW. Co-developed code OMS to generate scaling laws from on multiple, coupled, non-linear partial differential equations. Received AWS Silver Quill award for review paper on welding for aerospace applications.

Graduate Research Assistant, Department of Materials Science and Engineering 1994 – 1999
Established the foundation for scaling-modeling for analysis of complex, multicoupled materials processes. Invented and patented joining process using semi-solid metals. Collaborated with first additive manufacturing technology for fully dense metals using semi-solid processing, also patented. Investigated the root cause of high-speed welding defects and developed the mathematical and physical foundations for a quantitative criterion.

Teaching Assistant, 3.081 Materials Structure Laboratory Spring 1997
Taught recitations, graded, and developed original content. Course included heat treatments, arc welding, optical and electron microscopy, and mechanical testing of metals and ceramics.

Tenaris / CINI Campana, Argentina
Junior Research Engineer 1991 – 1993
Led a team of manufacturing and research engineers working on tribology aspects of manufacturing and performance of steel tubing for the oil industry. Designed a ring-on-disc tribological tester used for the development of Tenaris Dopeless Technology. Tester is still currently in use after more than 30 years. Completed year-long training program in management skills for high potential young professionals (IESE Business School, Buenos Aires).

University of Buenos Aires (UBA) Buenos Aires, Argentina
Teaching Assistant, 41.05 Calculus III 1987 – 1992
Taught recitations of most advanced course offered by the Department of Mathematics at the School of Engineering of UBA. The two-term curriculum included partial differential equations, asymptotic behavior, improper integrals, Fourier and Laplace transforms, and complex calculus. Led School of Engineering student team to third place in Argentina's nationwide university-level mathematical competition, ahead of the Department of Mathematics at the School of Science (1989).

Research Assistant (Department of Mechanical Engineering) 1988 – 1991
Focus on small wind turbines for remote dwellings. Developed a stability criterion for non-linear mechanical control of speed. Assembled a 1kW wind turbine on site. Developed a numerical scheme for the prediction of wind-turbine dynamic behavior which included dry friction and contact. Research carried at Argentine Navy Research Labs (SENID).

Academic Awards

2022 AWS William Spraragen Award to the best paper published in the Welding Journal Research Supplement ("Effect of the Heating Rate on Austenite Formation in Low Carbon Microalloyed Steels". Welding Journal, 2021. 100 (10): p. 338s-347s).
2016 CWA Michael N. Vucinich Award, to an individual who has "done the most to advance the science, technology and application of arc welding in Canada"
2016 CWA Gold Medal for best presentation at CWA Annual meeting the year before.
2016 AWS Charles H. Jennings Award, for most valuable contribution to the welding literature ("Primary Chromium Carbide Fraction Control with Variable Polarity SAW." Welding Journal. 94(1). pp. 1s-7s)
2015 AWS Warren F. Savage Memorial Award to the "best paper on metallurgical principles related to welding" published in the Welding Journal Research Supplement ("Characterization of High Strength Weld Metal Containing Mg-Bearing Inclusions." Welding Journal, 2014. 93 (1): p. 15s-22s.)
2015 Undergraduate Research Initiative University of Alberta Award for Outstanding Mentorship in Undergraduate Research & Creative Activities for "(having) gone above and beyond to help support the development of undergraduate researchers."
2014 ASM M. Brian Ives Lectureship Award to "an individual who has made distinguished and significant contributions to the Canadian materials community."
2014 AWS Fellow for a "career of significant achievements in the technical and research arenas."
2014 AWS Adams Memorial Membership Award for "outstanding teaching activities in undergraduate and postgraduate engineering."
2014 University of Alberta Student Union Award for Leadership in Undergraduate Teaching for "excellence in teaching by faculty members who make outstanding contributions in their roles as undergraduate instructors."
2013 CWA Fellowship Award for "exemplary reputation and service to advancements of welding sciences, technology application, research, education, publication of papers, books, journal articles and peer recognition."
2013 AWS William Irrgang Memorial Award "to the individual who had done the most to enhance AWS's goal of advancing the Science and Technology of Welding over the last five years."

2013 AWS Image of Welding: Educational Facility Award to “an educational facility that conducts welding education or training that has demonstrated an exemplary commitment to furthering the Image of welding”
2012 IIW Kenneth Easterling Best Paper Award to “the best contribution made over the three years proceeding on the advancement of knowledge or practice in respect of mathematical modelling of weld phenomena.” (OMS: A Computer Algorithm to Develop Closed-Form Solutions to Multicoupled, Multiphysics Problems in Proceedings of 10th International Seminar Numerical Analysis of Weldability, p. 219-254. Graz-Seggau, Austria)
2011 AWS William Spraragen Award to the best paper published in the Welding Journal Research Supplement (“A New Method for the Design of Welding Consumables”. Welding Journal, 2010. 89 (10): p. 201s-209s).
2006 NSF CAREER Award “NSF’s most prestigious awards in support of junior faculty who exemplify the role of teacher-scholars through outstanding research, excellent education and the integration of education and research”
2006 TMS Early Career Faculty Fellow Award, Certificate of Honorable Mention
2004 AWS Charles H. Jennings Award, for most valuable contribution to the welding literature (“Penetration and Defect Formation in High Current Arc Welding." Welding Journal. 82(10). pp. 296s-306s)
2004 Exponent/Failure Analysis Excellence Award
2002 AWS Silver Quill Award, for best original welding paper in non-welding journal (“Welding Processes for Aeronautics”. Advanced Materials & Processes, 2001. 159 (5): p. 39-43).
1997 Sigma Xi honor society, MIT chapter
1996 Alpha Sigma Mu honor society, MIT chapter
1994 First Place, Bridge Design Contest, MIT
1993 MIT Rocca Fellowship
1993 CONICET (Argentine National Research Council) Fellowship
1989 Argentine Construction Chamber Excellence Award (also in 1991)
1990 Argentine-American Cultural Institute Scholarship (also in 1991)
1988 University of Buenos Aires Research Fellowship, for studies on Wind Energy (also in 1989, 1990, 1991)
1984 First Place, “Elab” nationwide computer programming competition in Argentina involving more than 2,500 entries

Teaching

University of Alberta

Q221: Overall, this instructor was excellent. For Engineering, 50% percentile is 4.3, 75% percentile is 4.6.

Term	Course		% taught	Size	Median Q221
W22	CHE 314	Heat Transfer	100%	44	4.9
F21	MATE 481/681	Fund./Adv. Weld.	100%	23	5.0
W21	CHE 314	Heat Transfer	100%	58	N/A
F20	MATE 481	Fund. Weld.	100%	18	4.8
	MATE 681	Adv. Weld.	100%	4	N/A
W20	CHE 314	Heat Transfer	100%	58	N/A
F19	MATE 481	Fund. Weld.	100%	7	4.9
	MATE 681	Adv. Weld.	100%	13	4.8
	CME481	Colloquium I	100%	10	4.9
W19	CHE 314	Heat Transfer	100%	66	4.8
F18	MATE 466	Fund. Weld	100%	18	5.0
	MATE 630	Adv. Weld.	100%	4	N/A
W18	CHE 314	Heat Transfer	100%	63	4.6
F17	MATE 466	Fund. Weld	100%	21	4.7
	MATE 630	Adv. Weld.	100%	5	N/A
W17	CHE 314	Heat Transfer	100%	118	4.6
F16	MATE 466	Fund. Weld	100%	4	N/A
	MATE 630	Adv. Weld.	100%	4	N/A
W15	CHE 314	Heat Transfer	100%	106	2.6
	MATE 466	Adv. Manuf.	50%	13	4.4
	MATE 630	Adv. Manuf.	50%	8	4.4
	CME 483	Colloquium II	100%	22	4.4
F14	MATE 466	Fund. Weld	50%	30	4.1
	MATE 630	Adv. Weld.	50%	2	N/A
W14	CHE 314	Heat Transfer	100%	69	3.8
F13	MATE 466	Fund. Weld.	100%	26	4.7
	MATE 630	Adv. Weld.	100%	6	4.7
W13	MATE 202	Mat. Sci. II	100%	108	4.6
F12	MATE 630	Adv. Weld.	100%	9	5.0
	CME 481	Colloquium I	100%	24	4.9
W 12	MATE 202	Mat. Sci. II	100%	68	4.6
	CME 481	Colloquium I	100%	26	4.8
F 11	MATE 630	Adv. Weld.	100%	13	4.7
W 11	MATE 630	Weld. Met.	50%	11	4.3
	MATE 454	Weld. Met.	50%	11	4.0
	MATE 202	Mat. Sci. II	100%	82	4.6
F 10	MATE 335	Phase Trans I	50%	52	4.2
W 10	MATE 630	Weld. Met.	50%	12	3.9
	MATE 454	Weld. Met.	50%	12	4.7
	CME 483	Colloquium II	100%	26	4.8

Colorado School of Mines

Q14: Overall would you consider this instructor A-Superior, C-Average, E-Poor

Term	Course		% taught	Size	Rating Q14	Dept. Average
F 08	MTGN 587	Phys. of Weld	100%	5	N/A	N/A
	MTGN 300/2	Foundry Met.	100%	11	3.8	3.0
S 08	MTGN 548	Phase Trans	100%	11	3.0	3.2
	MTGN 300/2	Foundry Met.	100%	13	3.9	3.2
F 07	MTGN 300/2	Foundry Met.	100%	14	3.1	2.9
	MTGN 499	Indep. Studies	100%	5	N/A	N/A
S 07	MTGN 556	Transp. Solids	100%	26	2.8	3.0
	MTGN 300/2	Foundry Met.	100%	14	3.3	3.0
	MTGN 499	Indep. Studies	100%	4	N/A	N/A
F 06	MTGN 300/2	Foundry Met.	100%	13	3.6	3.0
S 06	MTGN 548	Phase Trans	100%	18	2.7	3.2
	MTGN 300/2	Foundry Met.	100%	18	2.4	3.2
	MTGN 498	Intr. Foundry	100%	6	N/A	N/A
F 05	MTGN 300/2	Foundry Met.	100%	18	3.0	3.3
S 05	MTGN 556	Transp. Solids	100%	11	3.4	3.1
F 06	MTGN 498	Adv. Casting	100%	6	N/A	N/A

Funding

Research Grants & Contracts

Total: $6.48M

Granting Agency	Company	Title	Date Start	Date End	Total
WD/PrairiesCan	Apollo Clad Group Six Universe Machine Xiris Inc Liburdi IRISNDT Alberta Laser Cladding Laserline Inc Oerlikon Metco	C:CLAD Cluster Development	22/09	27/03	$1,840,500
NSERC Alliance	Amsted Rail Apollo Clad	Developing an innovative, cost and quality advantageous approach to manufacturing and repair railroad wheels	20/01	26/05	$707,100
NSERC Alliance	Evraz NA	ERW Bondline Flaw Formation Mechanisms	22/12	24/11	$50,000
CWB-WF	CWB Welding Foundation	Program Support Fund	21/09	24/08	$125,000
CWB-WF	CWB Welding Foundation	Student Support Fund	21/09	24/08	$75,000
NSERC Discovery		Discovery of Governing Laws in Real Multiphysics, Multicoupled Systems in Materials Processing	18/04	24/03	$230,000
GCCIR	Group Six Technologies DURUM, BTU Brandenburg-Cottbus	DynaDur – Development and deposition of highly durable coating for dynamic-abrasive environments	21/04	24/03	$30,000
Industrial Research	Lincoln Electric	Parametric Optimization Study Aluminum Welding	22/09	23/08	$60,000
Industrial Research	AltaSteel	Developing a quantitative model of hydrogen through the complete production cycle of heat-treated, high-carbon grinding and SAG product	20/09	22/09	$98,000
UofA ISWSP		Modeling of heat transfer in welding	22/05	22/08	$4,000
MITACS Accelerate	Apollo Machine	Modelling and optimization of catchment efficiency in coaxial laser cladding	21/06	22/05	$55,000
NSERC CRD	Group Six Technologies	A novel laser-assisted hot-wire metal deposition technique for extreme applications (PI Rafiq Ahmad)	19/06	22/05	$158,400
CWB-WF	CWB Welding Foundation	Applied Research Fund	17/01	22/01	$75,000

MITACS Accelerate	Lincoln Electric Canada	Welding, brazing, consumables, GMAW, copper alloys, aluminum alloys, deposition rate, solidification properties	21/06	21/11	$15,000
NSERC CRD	CWB WF LJ Welding Weldco	Structure-Processing Relationships for Welding New Steels	19/09	21/08	$499,680
MITACS Accelerate	Xiris Inc.	Comparative analysis of real-time weld quality monitoring camera systems working in visible, IR and NIR spectra	21/03	21/06	$15,000
MITACS Accelerate	Xiris Inc.	Comparative analysis of real-time weld quality monitoring camera systems working in visible, IR and NIR spectra	20/09	20/12	$30,000
Government of Alberta (CARES)	Tackpoint Ltd.	Advancing Automation in the Alberta Resource Based Industry	18/01	19/12	$68,000
NSERC CRD	Enbridge GoldakTec MSC Software	Mechanical and Metallurgical Implications of Non-Ideal Geometry in Circumferential Pipeline Welds	16/12	19/09	$240,000
UofA ISWEP		ISWEP Summer Internship Developing Multimedia Teaching Content	19/05	19/08	$8,000
CWB-WF	Frank Roberts & Associates	Reinforcing Steels	18/10	19/04	$39,000
NSERC Discovery		Innovation in Materials Processing using Synthesis and Generalization of Multiphysics, Multicoupled Systems	14/04	19/03	$145,000
Government of Alberta		Manufacturing Megatrends	18/01	18/12	$45,000
NSERC USRA		Energy balance of a molten metal droplet surrounded by thermal plasma	18/05	18/08	$4,500
NSERC CRD	Apollo Clad	Heat and Mass Transfer Aspects of laser Deposition of Ni/WC Wear Resistant Metal Matrix Composites	14/08	18/07	$252,000
NSERC CRD	Wilkinson Steel	Processing and Microstructural Development of Wear Protection Coatings Based in the Fe-Cr-C System	14/11	18/03	$231,000
MITACS Accelerate	Group Six Technologies	Numerical Modelling of Laser Cladding Technology using Pastes and Tapes	17/09	18/02	$30,000
GCCIR	Group Six Technologies	New consumables for wear resistant overlays using pastes and tapes	17/09	17/12	$36,000
Industrial Research	Indalco Alloys	Parametric Optimization Study Aluminum Welding	16/03	16/12	$25,000

Industrial Research	CLAC	Multimedia-enabled Welding Helmet for Teaching and Welder Evaluation	15/05	16/12	$13,800
Government of Alberta		Trade Mission to Fabtech 2016, CWC 2016 Edmonton	16/01	16/12	$45,000
Government of Alberta		Trade Mission to Fabtech 2015	15/01	15/12	$30,000
NSERC Discovery		Innovation in Materials Processing Using Scaling Principles	09/04	15/03	$100,000
Auto21	Ford	Tailor-Welded Blank Manufacturing of Mg Parts	13/03	15/03	$66,400
Government of Alberta		Operating funds for Alberta-Germany technology collaboration	13/04	15/03	$30,000
Industrial Research	The Stoody Company	Wear Protective Overlays for Oil Sands Applications	12/04	14/09	$25,000
Productivity Alberta		Productivity Improvement Fund	13/10	14/09	$45,000
Industrial Research	The Stoody Company	Wear Protective Overlays for Oil Sands Applications	12/09	14/08	$30,000
NSERC USRA		Non-equilibrium isues in PTAW of Fe-Cr-C ternaries	14/05	14/08	$4,500
Government of Alberta		Operating funds for Alberta Metal Fabrication Innovation (AMFI) Program – Round 2	13/04	14/03	$20,000
NSERC Engage	Weld Management Solutions Inc.	Real-time monitoring of weld quality in flash-butt welding (PI Vinay Prasad)	13/05	14/01	$24,500
MITACS Accelerate	CASTI	Systematic catalog of welding defects	12/09	13/12	$22,000
NSERC CRD	Syncrude	Heat and Mass Transfer Phenomena in the Application of Wire-Based Nickel/Tungsten Carbide Overlays	10/04	13/08	$143,000
UofA ISWSP		Modeling of heat transfer in welding	13/05	13/08	$4,000
NSERC USRA		Torch Instrumentation in manual welding	13/05	13/08	$4,500
Government of Alberta		Operating funds for Alberta Metal Fabrication Innovation (AMFI) Program	11/04	13/03	$210,000
NSERC CRD	Hitachi Canada Industries	Deposition of abrasion-resistant Ni-WC metal matrix composites using Hot Wire TIG Welding	12/06	13/01	$100,000
NSERC CRD*	Syncrude	Mitigation of Embrittlement in Aged Stainless Steels (PI Adrian Gerlich)	10/04	12/06	$125,960
Government of Alberta		University/Industry Network for Welding and Joining	09/05	11/04	$95,000

NSERC Engage	H.C. Starck Canada Inc.	Development of Wire Consumables for Overlay Deposition by Hot-wire TIG Welding (PI Adrian Gerlich)	10/10	11/03	$20,950
Industrial Research	Hobart Brothers	Structure-Property Relationships in Advanced High Strength Steel Welds (PI Adrian Gerlich)	09/10	10/03	$19,134
Service Projects	Various	Industry Service Projects			$105,496

Equipment Grants
Total: $3.75M

Granting Agency	Company	Title	Date Start	Date End	Total
Industrial Partners		Donations/Consignment of Welding Equipment	10/05	99/12	$507,160
Canadian Foundation for Innovation (CFI)		Operational and Maintenance Funds	19/01	24/03	$50,000
PrairiesCan	Group Six Technologies KUKA Octopuz Weldco	Develop an advanced manufacturing system for automated repairs of heavy machine components	18/10	23/04	$1,060,000
Canadian Foundation for Innovation (CFI)		Development of Laser Processing Facility for Wear and Corrosion Protection Materials	15/01	19/12	$400,000
CWB-WF	Flir	Post-Secondary Equipment Fund - Thermal Camera for Welding Applications	16/09	17/03	$35,000
WD		Capital Equipment for Alberta Metal Fabrication Innovation (AMFI) Program	11/05	13/04	$1,500,000
NSERC RTI		A high accuracy analyzer for materials containing hydrogen, nitrogen, and oxygen	10/05	12/04	$111,273
NSERC RTI		High Temperature Mechanical Tester (PI Adrian Gerlich)	10/03	12/03	$87,212

At Colorado School of Mines
Total: US$1.11M

Funder	Title	Date Start	Date End	Total
DARPA	Spark Processing Synthesis of Reactive Materials Defense Advanced Projects Agency (PI Reed Ayers)	2009	2010	$200,000
Illinois Tool Works (ITW) / Miller Elelctric	Thermal Effects at the Contact Tip in GMAW	2008	2009	$70,000
National Science Foundation (NSF)	REU Supplement for CAREER Award. Division of Manufacturing and Industrial Innovation	2008	2009	$12,000
Tenaris/CONFAB (Brazil)	Filler Wire to Weld Tenaris Pipe	2007	2009	$190,000
Tenaris/TAMSA (Mexico)	Microstructural Analysis of Welding	2007	2008	$62,000

Cyco (Australia)	Durability of Titanium Equipment in the Manufacturing of Aluminum Matrix Composites	2007	2008	$10,000
National Science Foundation (NSF)	CAREER: Innovation in Materials Processing Using Scaling Principles. Division of Manufacturing and Industrial Innovation	2006	2010	$400,000
Novelis	Vertical Folds in Direct Chill Casting	2005	2008	$75,000
American Welding Society (AWS)	Modified GMAW for Spray Transfer with pure CO_2 shielding gas	2005	2008	$75,000
American Society for Non-Destructive Testing (ASNT)	Real-Time Determination of Solid Fraction in Semi-Solid Metal Melts Using Ultrasound Techniques	2005	2006	$15,000

Supervision

Postdoctoral Fellows Supervised

1. **Grams, M. R.**, *Catchment Efficiency in Laser Cladding*. Chemical and Materials Engineering, University of Alberta, 2021/7-2022/6. P.F. Mendez advisor. Now Laser Cladding R&D Engineer at Apollo Laser Clad, Edmonton AB.
2. **Havrylov, D.**, *Comparative Analysis of Real-Time Weld Quality Monitoring Camera Systems Working in Visible, IR and NIR Spectra*. Chemical and Materials Engineering, University of Alberta, 2021/1-2021/6. P.F. Mendez advisor. Now R&D Scientist, Xiris Automation, Burlington, ON.
3. **Ramirez, D.**, *Robotic Laser Cladding for Zero Programming Weld Repairs*. Chemical and Materials Engineering, University of Alberta, 2019/1-2022/7. P.F. Mendez advisor. Now Lead Robotics and Automation, Southern Alberta Institute of Technology (SAIT), Calgary AB.
4. **Kamyabi Gol, A.**, *Metallurgy of Hard Spots in Welded Rebar*. Chemical and Materials Enginering, University of Alberta, 2018/5-2018/9. P.F. Mendez advisor. Now R&D Engineer at Apollo Laser Clad, Edmonton AB.
5. **He, H.**, *Kinetics of Interlayers in Dissimilar Metal Weld Brazing* Chemical and Materials Engineering, University of Alberta, 2017/12-2019/11. P.F. Mendez advisor. Now Professor, School of Nuclear Equipment and Nuclear Engineering, Yantai University, China.
6. **Dapp, G.**, *Interdisciplinary Studies in Management of Research*. Chemical and Materials Engineering, University of Alberta, 2016/3-2018/6. P.F. Mendez advisor. Now Associate Director, Canadian Centre for Welding and Joining.
7. **Chapuis, J.**, *Calorimetry of Metal Transfer in Gas Metal Arc Welding*. Chemical and Materials Engineering, University of Alberta, 2011/8-2012/8. P.F. Mendez advisor. Now Manager of Welding Processes Unit at AREVA NP, France.
8. **Madeni, J. C.**, *Physical Modeling of SAW of Line Pipe Steel*. Metallurgical and Materials Engineering, Colorado School of Mines, 2005/8-2008/8. P.F. Mendez advisor. Now Manager Metallurgy & Advanced Materials at Johns Manville, Littleton CO.

PhD Theses Supervised

1. **Hintze Cesaro, A.**, *Austenite Formation in Low Carbon Microalloyed Pipeline Steels*. PhD thesis, Chemical and Materials Engineering, University of Alberta, 2022. P.F. Mendez advisor. Now Welding Research Engineer, EVRAZ North America, Regina, SK.
2. **Grams, M.**, *Predictive Expressions for Residual Stress and Distortion in Welding and Related Thermal Processes*. Ph.D. thesis, Chemical and Materials Engineering, University of Alberta, 2021. P.F. Mendez advisor. Now Research and Development Engineer, Apollo Machine & Welding, Leduc, AB.
3. **Havrylov, D.**, *Hydrodynamic Regimes in Autogenous Fusion Welding*. PhD thesis, Chemical and Materials Engineering, University of Alberta, 2021. P.F. Mendez advisor. Now Research and Development Scientist, Xiris Automation, Burlington ON.
4. **Lu, Y.**, *A Systematic Methodology to Develop Scaling Laws for Thermal Features of Temperature Field Induced by a Moving Heat Source*. PhD thesis, Chemical and Materials Engineering, University of Alberta, 2021. P.F. Mendez advisor. Now Research Engineer, United Imaging, Shanghai, China.

5. **Wang, Y.**, *Design Rules for Characteristics of Heat Flow in Welding on Thick Plates Using Asymptotics and Blending*. PhD thesis, Chemical and Materials Engineering, University of Alberta, 2021. P.F. Mendez advisor. Now Senior CAE Engineer, Tesla. Shanghai, China.
6. **Barnes, N.**, *Solidification Theory and Optimization of Chromium Carbide Overlays*. MSc thesis, Chemical and Materials Engineering, University of Alberta, 2017. P.F. Mendez advisor. Now Welding Engineer, Supreme Steel, Acheson, AB.
7. **Wood, G.**, *Heat and Mass Transfer Aspects of Coaxial Laser Cladding and Its Application to Nickel-Tungsten Carbide Alloys*. PhD thesis, Chemical and Materials Engineering, University of Alberta, 2017. P.F. Mendez advisor. Now Research and Development Engineer, Apollo Machine & Welding, Leduc, AB.
8. **Kamyabi Gol, A.**, *Quantification of Phase Transformations Using Calorimetry and Dilatometry*. PhD thesis, Chemical and Materials Engineering, University of Alberta, 2015. P.F. Mendez advisor. Now Research And Development Engineer, Apollo Machine & Welding, Leduc, AB.
9. **Guest, S. D.**, *Depositing Ni-Wc Wear Resistant Overlays with Hot-Wire Assist Technology*. PhD thesis, Chemical and Materials Engineering, University of Alberta, 2014. P.F. Mendez advisor. Now Senior Materials & Welding Specialist, BMT, Calgary, AB.
10. **Davis, L.**, *Wrinkling Phenomena to Explain Vertical Fold Defects in Dc-Cast Al-Mg4.5*. Ph.D. thesis, Metallurgical and Materials Engineering, Colorado School of Mines, 2009. P.F. Mendez advisor. Now Vice President of Operational Excellence at Wieland Group, Louisville, KY.
11. **Duman, Ü.**, *Modeling of Weld Penetration in High Productivity GTAW*. Ph.D. thesis, Metallurgical and Materials Engineering, Colorado School of Mines, 2009. P.F. Mendez advisor. Now Materials Technology Leader at Meta, Sunnyvale, CA.
12. **Soderstrom, E.**, *Gas Metal Arc Welding Electrode Heat and Mass Transfer Mechanisms*. Ph.D. thesis, Metallurgical and Materials Engineering, Colorado School of Mines, 2009. P.F. Mendez advisor. Now Principal Engineer at Collins Aerospace, Rockford, IL.
13. **Tordonato, D. S.**, *A Novel Approach to the Design of Welding Consumables Using Computational and Physical Models*. Ph.D. thesis, Metallurgical and Materials Engineering, Colorado School of Mines, 2008. P.F. Mendez advisor. Now Materials Engineer, US Bureau of Reclamation, Denver, CO.

MSc Theses Supervised

1. **Rojas Muñoz, D. I.**, *Considering a Temperature Gradient near the Pin for a Friction Stir Welding (FSW) Thermophysical Model Using Scaling Analysis*. MSc thesis, Mechanical Engineering, University of Chile, 2023. P.F. Mendez advisor.
2. **Segovia Pizarro, G. E.**, *Weldpool Recognition Using Deep Learning in the Laser Cladding Process*. MSc thesis, Mechanical Engineering, University of Chile, 2023. P.F. Mendez advisor.
3. **Romero Campos, J. P.**, *Machine Learning Applied to Welding Discontinuity Detection and Classification on Multipass Wire Feed Flux-Cored Arc Welding*. MSc thesis, Mechanical Engineering, University of Chile, 2022. P.F. Mendez advisor.
4. **González Pérez, I. I.**, *Welding Droplet Segmentation Using Deep Learning*. MSc thesis, Mechanical Engineering, University of Chile, 2021. P.F. Mendez and V. Meruane Naranjo advisors.
5. **Bell, K.**, *Full-Scale Testing of the Fatigue Life of Laser Additive Manufacturing Repaired Alloy Steel Components*. MSc thesis, Chemical and Materials Engineering, University of Alberta, 2017. P.F. Mendez advisor.
6. **Duncan, S.**, *Evaluation of Rock Particle Media Shape Contribution to Abrasive Wear*. MSc thesis, Mining Engineering, University of Alberta, 2016. P.F. Mendez and T. Joseph advisors.
7. **McIntosh, C.**, *Experimental Measurements of Fall Voltages and Droplet Temperature in GMAW*. MSc thesis, Chemical and Materials Engineering, University of Alberta, 2016. P.F. Mendez advisor.
8. **Sengupta, V.**, *High Speed Videography of Submereged Arc Welding*. MSc thesis, Chemical and Materials Engineering, University of Alberta, 2016. P.F. Mendez advisor.
9. **Tsui, J.**, *Modeling and Verification of FSW in Aluminum and Magnesium*. MSc thesis, Chemical and Materials Engineering, University of Alberta, 2016. P.F. Mendez advisor.
10. **Borle, S.**, *Microstructural Characterization of Chromium Carbide Overlays and a Study of Alternative Welding Processes for Industrial Wear Applications*. MSc thesis, Chemical and Materials Engineering, University of Alberta, 2014. P.F. Mendez advisor.

11. **Gajapathi, S. S.**, *Heat Transfer Analysis of Microwelding Using Tuned Electron Beam*. MSc thesis, Mechanical Engineering, University of Alberta, 2011. P.F. Mendez and S.K. Mitra advisors.
12. **Scott, K. M.**, *Heat Transfer and Calorimetry of Tubular Ni/WC Wires Deposited with GMAW*. MSc thesis, Chemical and Materials Engineering, University of Alberta, 2011. P.F. Mendez advisor.
13. **Gibbs, J. W.**, *Quantification of Phase Transformations Using Cooling Curves*. M.S. thesis, Metallurgical and Materials Engineering, Colorado School of Mines, 2009. P.F. Mendez and M.J. Kaufman advisors.
14. **Tello, K. E.**, *Coupled Model of Heat Transfer and Plastic Deformation for Friction Stir Welding Using Scaling Analysis*. M.S. thesis, Metallurgical and Materials Engineering, Colorado School of Mines, 2009. P.F. Mendez advisor.
15. **Soderstrom, E.**, *Influence of Ar-CO2 Mixtures and Thin Electrodes on Metal Transfer in Gas Metal Arc Welding*. M.S. thesis, Metallurgical and Materials Engineering, Colorado School of Mines, 2007. P.F. Mendez advisor.
16. **Nikou, V.**, *Welded Repair and Maintenance in the Space Environment*. S.M. thesis, Department of Materials Science and Engineering, Massachusetts Institute of Technology, 2003. T.W. Eagar, K. Masubuchi, and P.F. Mendez advisors.

MEng. Theses Supervised

1. **Panja, P.**, *Experimental Data Analysis of Significant Quenching Parameters*. M.Eng. thesis, Chemical and Materials Engineering, University of Alberta, 2023. P.F. Mendez advisor.
2. **Ma, R.**, *Survey and Validation of Models to Predict Hardness after Tempering of Steel Weldments*. M.Eng. thesis, Chemical and Materials Engineering, University of Alberta, 2020. P.F. Mendez advisor.
3. **Patel, V. R.**, *Survey and Generalization of Industrial Welding Procedures*. M.Eng. thesis, Chemical and Materials Engineering, University of Alberta, 2020. P.F. Mendez advisor.
4. **Singh, G.**, *The Analysis of Mechanical Properties of the Welding Zone for Submerged Arc Welding and Electric Resistance Welding of High Strength Low Alloy Linepipe: A Review*. Chemical and Materials Engineering, University of Alberta, 2020. P.F. Mendez advisor.
5. **Bell, M.**, *H2s Damage: Case Studies*. M.Eng. thesis, Chemical and Materials Engineering, University of Alberta, 2018. P.F. Mendez advisor.
6. **Jones, D.**, *Deposition Rate in Arc Welding*. Chemical and Materials Engineering, University of Alberta, 2018. P.F. Mendez advisor.
7. **Choi, L.**, *Outcome of Welding Parameters on Tungsten Carbide - Metal Matrix Composites Produced by GMAW*. MEng thesis, Chemical and Materials Engineering, University of Alberta, 2015. P.F. Mendez advisor.
8. **Narayanan, V.**, *Issues with in-Service or Repair Welding of Aged Pipelines - a Literature Review*. MEng thesis, Chemical and Materials Engineering, University of Alberta, 2014. P.F. Mendez advisor.
9. **Rollings, D.-a.**, *Characterization of Large Globules Present in Chromium Carbide Overlays Applied by Submerged Arc Welding*. MEng thesis, Chemical and Materials Engineering, University of Alberta, 2013. P.F. Mendez advisor.
10. **Nadeem, S.**, *Literature Review on Overlay Welding by Using Fe-Based Materials*. Chemical and Materials Engineering, University of Alberta, 2012. P.F. Mendez advisor.

BS Theses Supervised

1. **González Bisquertt, C.**, *Residual Stress and Strain Models in Pipeline Welding Considering Misaligned Joints*. Mechanical Engineering, University of Chile, 2022. P.F. Mendez advisor.
2. **Pérez Neira, T. A. S.**, *Modelo Acoplado De Transferencia De Calor Y Deformación Plástica En Soldadura Por Fricción-Agitación (FSW)*. Mechanical Engineering, University of Chile, 2022. P.F. Mendez advisor.
3. **Ramírez Vidal, F. J.**, *Levantamiento De Información De Modelos Predictivos De Distorsión Angular*. Mechanical Engineering, University of Chile, 2022. P.F. Mendez advisor.
4. **Urrutia Unda, M. E.**, *Evaluation of an Analytical Model of Friction Stir Welding*. Mechanical Engineering, University of Chile, 2022. P.F. Mendez advisor.
5. **Núñez Sánchez, V. A.**, *Energy Balance in Gas Metal Arc Welding*. Mechanical Engineering, University of Chile, 2021. P.F. Mendez advisor.
6. **Sacco Hawas, S.**, *Estudio De La Física Presente En Soldadura De Plasma*. Mech. Eng. thesis, Mechanical Enginering, University of Chile, 2020. P.F. Mendez advisor.

7. **Apaoblaza, D.**, *Estudio Del Comportamiento Del Plasma Termal a Presión Atmosférica Aplicado a Soldadura*. Mech. Eng. thesis, Mechanical Engineering, University of Chile, 2018. P.F. Mendez advisor.
8. **Vergara, B.**, *Modelamiento Acoplado Térmico Y De Deformación En Soldadura Por Fricción-Agitación*. Mech. Eng. thesis, Mechanical Engineering, University of Chile, 2017. P.F. Mendez advisor.
9. **Lopetegi, M.**, *Design and Experimental Production of X80 Microalloyed Filler Metal*. B.S. thesis, Department of Industrial Mechanical Engineering, Mondragón University, 2007. P.F. Mendez advisor.
10. **Sarasola, G.**, *Consumables Development for High Strength Steels in Arc Weldments*. B.S. thesis, Department of Industrial Mechanical Engineering, Mondragón University, 2006. P.F. Mendez advisor.

Service

Professional Memberships

- Professional Engineer, APEGA since 2013
- Professional Mechanical Engineer with federal jurisdiction in Argentina since 2005
- Member of CWA, AWS, ASM, ASME, CSME, ASEE

Editorial / Evaluator / Reviewer

Editorial

- Principal Reviewer Welding Journal (2005-current)
- Board of Review of AIST Transactions (Assoc. for Iron & Steel Technol., 2008-current)
- Associate Editor Journal of Manufacturing Processes (2014-current)
- Editorial Board of Science and Technology of Welding and Joining (2010-2015)
- Principal Reviewer Welding in the World (1/2012-12/2013)
- Guest editor for special issue on Materials Informatics in "Statistical Analysis and Data Mining" (with K. Rajan, 2008)

Evaluator

- Österreichische Forschungsförderungsgesellschaft mbH (Austrian Research Promotion Agency)

Reviewer

- Journals: Acta Materialia, Metallurgical and Materials Transactions A/B, Journal of Physics D: Applied Physics, Journal of Heat Transfer, International Journal of Heat and Mass Transfer, • Journal of Applied Mechanics, Materials Science and Enginering A, International Materials Reviews, Journal of Materials Processing Technology, Journal of Engineering Materials and Technology, Surface Coatings and Technology, Modelling and Simulation in Materials Science and Engineering, Engineering Computations, Statistical Analysis and Data Mining, Ocean Engineering, IEEE Transactions on Components and Packaging Technologies
- Funding Agencies: NSERC, MITACS, NSF (USA), CIMAT (Chile), International Copper Association

Program Chair

- Welding for the Hydrogen Economy, CWBA Seminar, April 28, 2023. Edmonton, AB.
- Welding for the Nuclear Industry, CWBA Seminar, June 27, 2022. Edmonton, AB.
- ASME section IX / ASME B31.3, CWBA Seminar, June 8-10, 2021. Edmonton, AB.
- Recubrimientos de Soldadura para el Desgaste, UTN Seminar, Argentina, Oct 21, 2021
- Getting the Welds you Need and Want, CWBA Seminar, March 3, 2020. Edmonton, AB.
- Structural Bolting, CWBA Seminar, March 4, 2020. Edmonton, AB.
- Design of Welded Connections: Steel Weldments, CWBA Seminar, May 9-10, 2019. Calgary, AB.
- Design of Welded Connections: Steel Structures, CWBA Seminar, May 6-7, 2019. Edmonton, AB.
- Tools for Quality, Productivity, and Economics of Welding, CWBA Seminar, June 14, 2018. Edmonton
- Design of Welded Structures, CWA Seminar, June 14-15, 2017. Edmonton, AB.
- Bringing Mathematical Solutions to the Engineers Drafting Table. Canadian Applied and Industrial Mathematics Society (CAIMS) Minisymposium, June 26, 2016.
- Structural Steel Welding, CWA Seminar, June 16, 2016. Edmonton, AB.
- Integrity of Pipelines Seminar, CWA Seminar, May 28, 2015. Edmonton, AB.
- Duplex and Super Duplex Stainless Steels, CWA Seminar, May 27, 2014. Edmonton, AB.

- Weld Overlays for Wear Protection, CWA Seminar, May 16, 2013. Edmonton, AB.
- Numerical, Mathematical, and Physical Modeling Tools for Materials Processes, TMS/MS&T Symposium, Nov. 19-20, 2007.

Commitees and Other Service

University of Alberta

- Weldco/Industry Chair in Welding and Joining (2009–present)
- Founder and Faculty Advisor for AWS and CWA Student Chapters (2010-present, both are the first University Chapters in Canada).
- UofA Laser safety committee (2023–present)
- Host CCWJ Open Labs, which happen regularly four times a year exposing 40 students to hands-on welding each time (2010-present).
- Host section of Engineering Expo which happens every year exposing ~150 high school students/parents per event (2010-present).
- Host section of UofA Open House which happens every year exposing ~150 high school students/parents per event (2010-present).
- D.B. Robinson Distinguished Speaker Series (9/2010-12/2013)
- Judge for Great Northern Concrete Toboggan Race hosted at UofA (2011)
- Assisted students from the formula SAE team with their welding needs

Colorado School of Mines

- "Key Professor" Foundry Educational Foundation (2006-2009)
- Founder and Faculty Advisor for Welding Club and Friday Free Pour Club at CSM, an Informal Science Education approach where students are exposed to engineering aspects of welding and casting outside a classroom setting (2006).
- Founder of the Undergraduate Research Leaders Scholarship, the first funded school-wide undergraduate research program at CSM (2006).
- Head of Selection committee for Undergraduate Research Leaders Scholarship (URLS) scholarship (2006-2007)
- CSM/MME Undergraduate Affairs Committee (2004-2008)
- Reported to the Technology Transfer Advisory Board on the development of a Business Plan Competition (2007)
- Host in Discover CSM (two 3h metal casting demos for ~50 students)
- Host in CSM 101 (1h welding and metal casting ~50 students, three days per semester, 2004-2008)
- Host in Preview Mines (4h metal casting demos for ~50 students every semester, 2004-2008)
- Hosted foundry demo for Creighton Middle School (4h metal casting demo for ~30 students, 2007)
- Hosted foundry demo for President Scoggins (2007)

Professional Societies

- AWS A5G Subcommittee on Hard Facing Filler Metals– (7/2014-current)
- AWS EC Education Committee, Higher Education – Engineering Subcommittee (2/2014-current)
- ASME HTD K-15 Committee on Transport Phenomena in Manufacturing and Materials Processing (2009-current)
- CWA Edmonton Executive Committee (2011–current)
- AWS Alberta Section Education Co-Chair (2009-current)
- AWS Welding Research & Development Committee (2008-current)
- AWS A9 Computational Weld Mechanics Committee (2007-current)
- AWS Technical Papers Committee (2005-current)
- TMS Computational Materials Science and Engineering Committee (2005-2009)
- TMS Process Modeling Analysis and Control Committee (2005-2009)

Professional Leadership

- Co-Founder, and Executive Committee Member (International Liaison) of the Chilean Society for Welding and Additive Manufacturing (2023-present)
- Canadian Delegate to the IIW (2011-present)

- Co-Founder of MicroAI, a startup which developed one of the first apps able to classify metal microstructures by morphology, including the complef microstructures from a heat affected zone in welding (2020-2022)
- IIW Technical Management Board (2015-2018)
- Founder, Alberta Metal Fabrication Innovation (AMFI) Program: This is a $3.85M initiative in partnership with the Government of Alberta, Federal Government (Western Economic Development, now PrairiesCan), and Alberta Innovates. This initiative aims to revolutionize the metal fabrication industry in Alberta by providing engineering training and assistance to small and medium enterprises on insightful choices of the latest available productivity technologies. Ongoing program, started in Jan 1, 2011.
- Board of Directors, MaJIC (Materials Joining Innovation Centre, Ontario), 2016-present.
- Scientific Committee, International Seminar Numerical Analysis of Weldability, 2012, 2015, 2018, 2022 Seggau, Austria
- Scientific Advisory Committee, International Institute of Welding (IIW) Annual Conference, July 2015, Helsinki, Finland.
- International Advisory Committee, 2nd International Symposium on Computer-Aided Welding Engineering, Aug. 2012, Jinan, China.
- AWS Roundtable in Welding Education (Sep. 2011). Only University-level educator invited to participate in this high-level roundtable with 11 corporate directors and 3 directors of other educational and training institutions. Conclusions were published in "Strategies for Developing Tomorrows Welding Professionals" by AWS in 2012.
- Government of Alberta (Alberta Finance and Enterprise) Manufacturing Competitiveness and Productivity Roundtable (Nov. 2010). Represented university education at a roundtable including an MLA.
- Co-founder and first President of the Rocca Fellowship Association at MIT (1998)
- President of Edgerton House (MIT most modern graduate residence. Elected 1996, re-elected uncontested 1997)
- Treasurer, Argentinean Club at MIT (1997, 1998)

Publications

(HQP supervised in **boldface**)

Papers in archival journals

1. **González Pérez, I.**, Meruane, V., and Mendez, P. F., *Deep-Learning Based Analysis of Metal-Transfer Images in GMAW Process.* Journal of Manufacturing Processes, 2023. **85**: p. 9-20.
2. Prisco, U., **Duman, U.**, and Mendez, P. F., *Scaling Modelling of Penetration in High Productivity Gas Tungsten Arc Welding.* Journal of Materials Processing Technology, 2023. **320** (118120): p. 1-16.
3. **Ramirez Rebollo, D. R.**, **Alam, S. I.**, and Mendez, P. F., *Zero Programming Robotics in Additive Manufacturing Repairs.* Journal of Advanced Joining Processes, 2023. **7**: p. 100145.
4. **Villarreal-Medina, R.**, Murphy, A. B., Méndez, P. F., and Ramírez-Argáez, M. A., *Heat Transfer Mechanisms in Arcs of Various Gases at Atmospheric Pressure.* Plasma Chemistry and Plasma Processing, 2023: p. 1-17.
5. Castro Cerda, F. M., Goulas, C., Jones, D., Kamyabi, A., Hamre, D., Méndez, P., and Wood, G., *Modelling the Laser Surface Hardening in a Ferrite and Pearlite Initial Microstructure.* Materials Science and Technology, 2023: p. 1-11.
6. Liu, H., Qi, B., Qi, Z., Zhu, Y., Guo, W., Yang, M., and Mendez, P. F., *Mechanism of Constriction in a High Frequency Pulsed Welding Arc.* International Journal of Heat and Mass Transfer, 2023. **211**: p. 124251.
7. **Grams, M.** and Mendez, P. F., *Effect of Compliance on Residual Stresses in Manufacturing with Moving Heat Sources.* Journal of Applied Mechanics, 2022. **89**: p. 021004.
8. **Wang, Y.** and Mendez, P. F., *Isotherm Penetration Depth under a Moving Gaussian Surface Heat Source on a Thick Substrate.* International Journal of Thermal Sciences, 2022. **172**: p. 107334.
9. **Yan, Z.**, **Scott, K. M.**, Chen, S., and Mendez, P., *Deposition Rate in GMAW of Er1100 and Er5183 Aluminum Alloys.* Welding Journal, 2022. **101** (11): p. 289s-292s.
10. **Grams, M. R.** and Mendez, P. F., *A General Expression for the Welding Tendon Force.* Journal of Manufacturing Science and Engineering, 2021. **143** (12): p. 121002-1, 121002-12.

11. **Grams, M. R.** and Mendez, P. F., *Scaling Analysis of the Thermal Stress Field Produced by a Moving Point Heat Source in a Thin Plate.* Journal of Applied Mechanics, 2021. **88**: p. 011001, 1-10.
12. **Hintze Cesaro, A.** and Mendez, P. F., *Effect of the Heating Rate on Austenite Formation in Low Carbon Microalloyed Steels.* Welding Journal, 2021. **100** (10): p. 338s-347s.
13. **Lu, Y.** and Mendez, P. F., *Characteristic Values of the Temperature Field Induced by a Moving Line Heat Source.* International Journal of Heat and Mass Transfer, 2021. **166**: p. 120671, 1-15.
14. Sendetskyi, O., Salomons, M., Mendez, P., and Fleischauer, M., *Conflat Cell for in-Situ Electrochemical X-Ray Studies of Lithium-Ion Battery Materials in Commercially-Relevant Conditions.* Journal of Applied Crystallography 2021. **54**: p. 1416-1423.
15. Molina, C., Araujo, A., **Bell, K.**, Mendez, P. F., and Chapetti, M., *Fatigue Life of Laser Additive Manufacturing Repaired Steel Component.* Engineering Fracture Mechanics, 2021. **241**: p. 107417, 1-11.
16. **Velázquez-Sánchez, A.**, **Delgado-Álvarez, A.**, Mendez, P. F., Murphy, A. B., and Ramírez-Argáez, M. A., *Dominant Heat Transfer Mechanisms in the GTAW Plasma Arc Column.* Plasma Chemistry and Plasma Processing, 2021. **41**: p. 1497-1515.
17. **Lu, Y.**, **Wang, Y.**, and Mendez, P. F., *Width of Thermal Features Induced by a 2-D Moving Heat Source.* International Journal of Heat and Mass Transfer, 2020. **156**: p. 119793, 1-14.
18. **Delgado-Álvarez, A.**, Mendez, P. F., Murphy, A. B., and Ramírez-Argáez, M. A., *Generalized Representation of Arc Shape, Arc Column Characteristics and Arc-Weld Pool Interactions for Dc Electric Arcs Burning in Monoatomic Gases.* Journal of Physics D: Applied Physics, 2020. **54** (5): p. 055001, 1-12.
19. **Delgado-Álvarez, A.**, Mendez, P. F., and Ramírez-Argáez, M. A., *Dimensionless Representation of the Column Characteristics and Weld Pool Interactions for a Dc Argon Arc.* Science and Technology of Welding and Joining, 2019.
20. **Sengupta, V.**, **Havrylov, D.**, and Mendez, P., *Physical Phenomena in the Weld Zone of Submerged Arc Welding—a Review.* Welding Journal, 2019. **98** (10): p. 283S-313S.
21. **Wang, Y.**, **Lu, Y.**, and Mendez, P. F., *Scaling Expressions of Characteristic Values for a Moving Point Heat Source in Steady State on a Semi-Infinite Solid.* International Journal of Heat and Mass Transfer, 2019. **135**: p. 1118-1129.
22. **He, H.**, Gou, W., Lin, S., Yang, C., and Mendez, P., *Gta Weld Brazing a Joint of Aluminum to Stainless Steel.* Welding Journal, 2019. **98** (12): p. 365s-378s.
23. **He, H.**, Gou, W., Wang, S., Hou, Y., Ma, C., and Mendez, P. F., *Kinetics of Intermetallic Compound Layers During Initial Period of Reaction between Mild Steel and Molten Aluminum.* International Journal of Materials Research, 2019. **110** (3): p. 194-201.
24. Mendez, P. F., **Wang, Y.**, and **Lu, Y.**, *Scaling Analysis of a Moving Point Heat Source in Steady-State on a Semi-Infinite Solid.* Journal of Heat Transfer, 2018. **140**: p. 081301-9.
25. **Barnes, N.**, Clark, S., Seetharaman, S., and Mendez, P. F., *Growth Mechanism of Primary Needles During the Solidification of Chromium Carbide Overlays.* Acta Materialia, 2018. **151**: p. 356-365.
26. **McIntosh, C.** and Mendez, P. F., *Experimental Measurements of Fall Voltages in Gas Metal Arc Welding.* Welding Journal, 2017. **96**: p. 121s-132s.
27. **McIntosh, C.** and Mendez, P. F., *Fall Voltages in Advanced Waveform Aluminum GMAW.* Welding Journal, 2017. **96**: p. 354s-366s.
28. **Sengupta, V.** and Mendez, P. F., *Effect of Current on Metal Transfer in SAW. Part 1: Dcep.* Welding Journal, 2017. **96**: p. 241s-249s.
29. **Sengupta, V.** and Mendez, P. F., *Effect of Current on Metal Transfer in SAW. Part 2: Ac.* Welding Journal, 2017. **96**: p. 271s-277s.
30. **Sengupta, V.** and Mendez, P. F., *Effect of Fluxes on Metal Transfer and Arc Length in SAW.* Welding Journal, 2017. **96**: p. 334s-353s.
31. **Barnes, N.**, **Borle, S.**, and Mendez, P. F., *Large Anomalous Features in the Microstructure of Chromium Carbide Weld Overlays.* Science And Technology Of Welding And Joining, 2017. **22** (7): p. 595-600.
32. **Kamyabi Gol, A.**, **Herath, D.**, and Mendez, P. F., *A Comparison of Common and New Methods to Determine Martensite Start Temperature Using a Dilatometer.* Canadian Metallurgical Quarterly, 2016. **56** (1): p. 85-93.
33. **McIntosh, C.**, **Chapuis, J.**, and Mendez, P. F., *Effect of Ar-CO2 Gas Blends on Droplet Temperature in GMAW.* Welding Journal, 2016. **95** (8): p. 273s-279s.

34. **Kamyabi-Gol, A.**, Clark, S. J., Gibbs, J. W., Seetharaman, S., and Mendez, P. F., *Quantification of Multiple Simultaneous Phase Transformations Using Dilation Curve Analysis (Dca).* Acta Materialia, 2016. **2016**: p. 231-240.
35. **Kamyabi-Gol, A.** and Mendez, P. F., *The Evolution of the Fraction of Individual Phases During a Simultaneous Multiphase Transformation from Time–Temperature Data.* Material Transactions A, 2015. **46A** (2): p. 622-638.
36. **Wood, G.** and Mendez, P. F., *Disaggregated Metal and Carbide Catchment Efficiencies in Laser Cladding of Nickel-Tungsten Carbide.* Welding Journal, 2015. **94** (11): p. 343s-350s.
37. **Barnes, N.**, Joseph, T., and Mendez, P. F., *Issues Associated with Welding and Surfacing of Large Mobile Mining Equipment for Use in Oil Sands Applications.* Science And Technology Of Welding And Joining, 2015. **20** (6): p. 483-493.
38. **Borle, S. D.**, **Gall, I. L.**, and Mendez, P. F., *Primary Chromium Carbide Fraction Control with Variable Polarity SAW.* Welding Journal, 2015. **94** (1): p. 1s-7s.
39. Mendez, P. F., Goett, G., and **Guest, S. D.**, *High-Speed Video of Metal Transfer in Submerged Arc Welding.* Welding Journal, 2015. **94** (10): p. 325s-332s.
40. **Gibbs, J. W.**, Schlachter, C., Mayr, P., **Kamyabi-Gol, A.**, and Mendez, P. F., *Cooling Curve Analysis as an Alternative to Dilatometry in Continuous Cooling Transformations.* Materials Transactions A, 2015. **46A** (1): p. 148-155.
41. **Wood, G.**, **Islam, S. A.**, and Mendez, P. F., *Calibrated Expressions for Welding and Their Application to Isotherm Width in a Thick Plate.* Soldagem & Inspecao, 2014. **19** (3): p. 212-220.
42. Gerlich, A. P., **Izadi, H.**, Bundy, J., and Mendez, P. F., *Characterization of High Strength Weld Metal Containing Mg-Bearing Inclusions.* Welding Journal, 2014. **93** (1): p. 15s-22s.
43. **Guest, S.**, **Chapuis, J.**, **Wood, G.**, and Mendez, P. F., *Non-Wetting Behavior of Tungsten Carbide Powders in a Nickel Weld Pool: A New Loss Mechanism in GMAW Overlays.* Science And Technology Of Welding And Joining, 2014. **19** (2): p. 133-141.
44. **Barnes, N.**, **Borle, S.**, **Dewar□, M.**, **Andreiuk, J.**, and Mendez□, P. F., *3d Microstructure Reconstruction of Chrome Carbide Weld Overlays.* Science And Technology Of Welding And Joining, 2014. **19** (8): p. 696-702.
45. Mendez, P. F., **Barnes, N.**, **Bell, K.**, **Borle, S.**, **Gajapathi, S. S.**, **Guest, S. D.**, **Izadi, H.**, **Gol, A. K.**, and **Wood, G.**, *Welding Processes for Wear Resistant Overlays.* Journal of Manufacturing Processes, 2014. **16**: p. 4-25.
46. Mendez, P. F. and Eagar, T. W., *Order of Magnitude Scaling: A Systematic Approach to Approximation and Asymptotic Scaling of Equations in Engineering.* Journal of Applied Mechanics, 2013. **80** (1): p. 011009-1 to 9.
47. **Gajapathi, S. S.**, Mitra, S. K., and Mendez, P. F., *Part 2: Application of Kanaya–Okayama Heat Source in Modelling Micro Electron Beam Welding.* Science and Technology of Welding and Joining, 2012. **17** (6): p. 435-440.
48. **Gajapathi, S. S.**, Mitra, S. K., and Mendez, P. F., *Part 1: Development of New Heat Source Model Applicable to Micro Electron Beam Welding.* Science And Technology Of Welding And Joining, 2012. **17** (6): p. 429-434.
49. Mendez, P. F., *Synthesis and Generalisation of Welding Fundamentals to Design New Welding Technologies: Status, Challenges and a Promising Approach.* Science And Technology Of Welding And Joining, 2011. **16** (4): p. 348-356.
50. **Lehnhoff, G.** and Mendez, P. F., *Scaling of Non-Linear Effects in Heat Transfer of a Continuously Fed Melting Wire.* International Journal of Heat and Mass Transfer, 2011. **54**: p. 2651-2660.
51. **Gajapathi, S. S.**, Mitra, S. K., and Mendez, P. F., *Controlling Heat Transfer in Micro Electron Beam Welding Using Volumetric Heating.* International Journal of Heat and Mass Transfer, 2011. **54** (25-26): p. 5545-5553.
52. **Soderstrom, E.**, **Scott, K. M.**, and Mendez, P. F., *Calorimetric Measurement of Droplet Temperature in GMAW.* Welding Journal, 2011. **90** (4): p. 77s-84s.
53. **Marcano, D.**, Mendez, P. F., **Gibbs, J. W.**, and Kannengiesser, T., *Martensite Fraction Determination Using Cooling Curve Analysis.* Solid State Phenomena, 2011. **172-174**: p. 221-226.
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3. Mendez, P. F. *Functions of Polarity, Pulsing, Gas, and Fluxes in Arc Welding.* Congreso Internacional de Soldadura. May 19-21, 2021. Junin, Argentina.
4. Mendez, P. F. *State of the Art Materials and Technologies for the Deposition of Wear Resistant Coatings.* Simposio Internacional de Soldadura. April 28-30, 2021. Buenos Aires, Argentina.
5. Mendez, P. F. *Design of Procedures in Welding and Additive Manufacturing of Metals.* AWS Annual Meeting. Sep. 20-23, 2021. Virtual.
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69. **Guest, S.**, Gerlich, A., and Mendez, P. F. *Depositing Ni-Wc Wear Resistant Coatings with Hot-Wire Assisted GTAW*. Fabtech/AWS Annual Meeting 2011. Chicago, IL.
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1. Mendez, P. F., *Order of Magnitude Scaling of Complex Engineering Problems, and Its Application to High Productivity Arc Welding*. Doctor of Philosophy dissertation, 1999. Massachusetts Institute of Technology, Department of Materials Science and Engineering, advisor.
2. Mendez, P. F., *Joining Metals Using Semi-Solid Slurries*. Master of Science dissertation, 1995. Massachusetts Institute of Technology, Department of Materials Science and Engineering, advisor.

Invited talks

1. Mendez, P.F. *Deep Physics and Concrete Needs. Welding as the Gateway of Science, Engineering, and Excitement.* in *Brian Ives Award Lecture Series*. Jun. 9, 2015. Calgary, AB.
2. Mendez, P.F. *Engineering as a Career Path.* in *Career Day, Grandview Heights* May 19, 2015, 2015. Edmonton, AB.
3. Mendez, P.F. *Modeling of welding for engineering applications.* in *Yonsei University*. Jul. 16, 2014. Seoul, Korea.
4. Mendez, P.F. *Transferrable Skills and Links with the Private Sector.* in *Joint CALDO-CRUCH Seminar*. May 14-16, 2014. Pucon, Chile.
5. Mendez, P.F. *Support Strategies for PhD Success.* in *Joint CALDO-CRUCH Seminar*. May 14-16, 2014. Pucon, Chile.
6. Mendez, P.F. *B.SC. Program in Materials Engineering.* in *Grant McEwan University*. Jan. 31, 2014. Edmonton, AB.
7. Mendez, P.F. *Modeling of welding for engineering applications.* in *Materials Science and Engineering*. Jul. 17, 2014. Seoul, Korea.
8. Mendez, P.F. *Modeling of welding for engineering applications.* in *Beijing University of Technology, Harbin Institute of Technology, Jiaoton University*. July 15, 16, 17, 2013. Beijing, Harbin, Xi'an.
9. Mendez, P.F. *The Joy of Welding, and why it is much deeper than it seems.* in *Materials Engineering Technical Society (METS)*. Mar. 14, 2013. Edmonton, AB.
10. Mendez, P.F., **Guest, S.D.**, and Gilbank, J. *A systematic approach to procedure development.* in *Calgary Lecture Series*. Mar. 17, 2013. Calgary, AB.
11. Mendez, P.F., **Guest, S.D.**, and Gilbank, J. *A systematic approach to procedure development.* in *Edmonton Lecture Series*. Feb. 19, 2013. Edmonton, AB.
12. Mendez, P.F. *Asymptotic Analysis of Coupled Phenomena in FSW.* in *Materials Science and Engineering*. Aug. 22, 2012. Jinan, China.
13. Mendez, P.F. *Humping in High Speed GTAW.* in *Materials Science and Engineering*. Aug. 21, 2012. Jinan, China.
14. Mendez, P.F. *Modeling using asymptotic scaling.* in *Laboratoire de Mécanique et Génie Civil de Montpellier*. Jul. 13, 2011. Montpellier, France.
15. Mendez, P.F. *Coupled heat transfer and deformation in FSW.* in *Materials Joining Group*. Jun. 9, 2011. Oak Ridge, TN.
16. Mendez, P.F. *Generalisation and communication of findings using scaling analysis.* in *Phase Transformations & Complex Properties Research Group*. July 7, 2011. Cambridge, UK.

17. Mendez, P.F. *NSERC Strategic Network on Pipeline Materials Reliability - Theme 5: Welding.* in *Mechanical Engineering*. April 8, 2011. Calgary, AB.
18. Mendez, P.F. and Gerlich, A.P. *The Canadian Centre for Welding and Joining.* in *NRC/IRAP Welding Productivity Improvement*. Feb. 9, 2011. Edmonton, AB.
19. Mendez, P.F. *Welding Research, Education and Technologies.* in *PeRMA Welding & Manufacturing Information Sharing Session*. Dec. 10, 2010. Grande Prairie, AB.
20. Mendez, P.F. *Scaling Techniques for Multiphysics, Multicoupled Engineering Problems.* in *Theoretical and Applied Mechanics Seminar, University of Alberta*. Nov. 25, 2009. Edmonton, AB.
21. Mendez, P.F. and Gerlich, A.P. *Canadian Centre for Welding and Joining. An interdepartmental research and educational effort.* in *CISC Alberta Region / University of Alberta Annual Meeting*. Apr. 20, 2009. Edmonton, AB.
22. Mendez, P.F. *Welding Innovation: Applications, fundamentals, and a modeling approach to link them.* in *Department of Chemical and Materials Engineering*. June 20, 2008. Edmonton, AB.
23. Mendez, P.F. *Scaling Laws in Manufacturing Involving Materials Processing.* in *Department of Mechanical Engineering*. April 25, 2008. Lexington, KY.
24. Mendez, P.F. *Scaling Laws in Friction Stir Welding and Other Multicoupled Problems.* in *Department of Mechanical Engineering*. April 14, 2008. Nashville, TN.
25. Mendez, P.F. *Modeling of Materials Processing Using Scaling Laws.* in *Department of Metallurgical and Materials Engineering*. November 29, 2007. Tuscaloosa, AL.
26. Mendez, P.F. *Seed Grants for Tech Transfer: MIT Example.* in *Technology Transfer Advisory Board*. September 7, 2007. Golden, CO.
27. Mendez, P.F. *Analysis of Manufacturing Processes Involving Molten Metals: Behavior of High Speed GTAW.* in *Department of Industrial, Welding, and Systems Engineering*. April 30, 2007. Columbus, OH.
28. Mendez, P.F. *Modeling of Welding and Materials Processes Using Scaling Techniques.* in *MST-6*. March 13, 2007. Los Alamos, NM.
29. Mendez, P.F. *Reconstructing Physical Laws from Statistical Data.* in *Department of Physics*. October 24, 2006. Golden, CO.
30. Mendez, P.F. *Parametric Modeling In Materials Processing.* in *Department of Metallurgical and Materials Engineering*. February 9, 2006. Golden, CO.
31. Mendez, P.F. *Innovations in Metallurgical Manufacturing: From concept to implementation.* in *Department of Industrial Engineering*. January 6, 2005. Los Angeles, CA.
32. Mendez, P.F. and Liu, S. *Consumable Development for High* in *Welding Department*. Sep. 14, 2005. Pindamonhangaba, Brazil.
33. Mendez, P.F. *Temas de avanzada en soldadura (Leading trends in welding).* Dec. 9, 2004. Ordizia, Spain.
34. Mendez, P.F. *Scaling Laws for Manufacturing, Design, and Beyond.* in *Department of Mechanical Engineering*. March 3, 2003. Austin, TX.
35. Mendez, P.F. *Tackling welding complexity: A new approach and new results.* in *Department of Metallurgical and Materials Engineering*. July, 2003. Golden, CO.
36. Mendez, P.F. *Order of Magnitude Scaling.* in *Similarity Mechanics Group, Institute for Statics and Dynamics of Aerospace Structures*. November, 2001. Stuttgart, Germany.
37. Mendez, P.F. *Novel Techniques for Understanding Complex Materials Processes.* in *Department of Materials Science and Engineering*. November, 2001. Cambridge, MA.
38. Mendez, P.F. *DOE/BES Work at the MIT Welding Laboratory.* in *Department of Energy, Basic Energy Sciences Review Meeting*. June, 2000. Cambridge, MA.
39. Mendez, P.F. *Estudios de Postgrado de Ingeniería en USA.* Dec. 21, 1999. Buernos Aires, Argentina.
40. Mendez, P.F. *Suppression of Defects in High-Productivity Arc Welding.* in *TMS Lecture Series*. October, 1999. Cambridge, MA.
41. Mendez, P.F. *Order of Magnitude Scaling of Complex Engineering Problems and its Application to High Productivity Arc Welding.* in *Department of Energy, Basic Energy Sciences Review Meeting*. June, 1999. Idaho Falls, ID.
42. Mendez, P.F. *Transport Phenomena in Welding.* in *3.55: Macroscopic Transport in Materials Processing*. July, 1998. Cambridge, MA.
43. Mendez, P.F. *Magnitude Scaling of Welding Problems.* in *Department of Energy, Basic Energy Sciences Review Meeting*. June, 1998. Cambridge, MA.

44. Mendez, P.F. *Metal Transfer in GMAW in the High Current Regime.* in *Department of Energy, Basic Energy Sciences Review Meeting*. June, 1997. Idaho Falls, ID.
45. Weiss, D., Mendez, P.F., and Eagar, T.W. *Simulation of metal transfer effects in GMAW.* in *Department of Energy, Basic Energy Sciences Review Meeting*. June, 1997. Idaho Falls, ID.
46. Mendez, P.F. *Mathematical Modeling of Heat and Fluid Flow Phenomena in Gas Metal Arc Welding.* in *Department of Energy, Basic Energy Sciences Review Meeting*. July, 1996. Cambridge, MA.

Poster presentations

1. Hiscocks, J., Knight, S., **Tsui, J.**, Diak, B., Daymond, M., Gerlich, A., and Mendez, P.F., *Friction Stir Welding of Magnesium Alloy Wheels*, in *AUTO 21 Annual Meeting*. 2014: Niagara Falls, ON.
2. Hiscocks, J., Knight, S., **Tsui, J.**, Diak, B., Daymond, M., Gerlich, A., and Mendez, P.F., *Friction Stir Welding of Magnesium Alloy Wheels*, in *AUTO 21 Annual Meeting*. 2013: Kingston, ON.
3. **Duman, U.** and Mendez, P.F., *Modeling of weld penetration in high velocity GTAW*, in *TMS Annual Meeting*. 2011, TMS: San Diego, CA.
4. **Roubidoux, J.A.** and Mendez, P.F., *Advanced Scaling Techniques for the Modeling of Materials Processing*, in *NSF Design, Service, and Manufacturing Grantees and Research Conference*. 2006: St. Louis, MO.
5. Mendez, P.F. and Eagar, T.W., *Suppression of Defects in High Productivity Welding*, in *Annual Materials Processing Center Research Review*. 1998: Cambridge, MA.
6. Mendez, P.F. and Eagar, T.W., *Modeling Penetration and Free Surface Depression During High Current Arc Welding*, in *Julian Szekely Memorial Symposium on Materials Processing/ 1997 TMS Fall Extraction and Processing Conference*. 1997, TMS: Cambridge, MA. p. 719.
7. Mendez, P.F. and Eagar, T.W., *Mathematical Modeling of Heat and Fluid Flow Phenomena in Gas Metal Arc Welding*, in *Annual Materials Processing Center Research Review*. 1996: Cambridge, MA.

Technical reports

1. Odriozola, I., Colombo, G., Mendez, P.F., Rapetti, J., and Herrera, R., *Utilización del Concepto de Pico Móvil para la Aplicación de Lubricantes a Base de Grafito (On the Use of Moving Nozzles for the Application of Graphite-Based Lubricants)*. Report # I 604/93, 1993. FUDETEC, Campana, Argentina.
2. Mendez, P.F. and Bastianon, R.A., *Regulación Centrífuga de Turbinas Eólicas (Centrifugal Governors on Wind Turbines)*. Report # 3, 1992. Universidad de Buenos Aires, Buenos Aires, Argentina.
3. Mendez, P.F. and Herrera, R., *Mecanismos de Desgaste (Wear Mechanisms)*. Report # I 1101/483-92, 1992. FUDETEC, Campana, Argentina.
4. Mendez, P.F. and Herrera, R., *Aplicabilidad del Lubricante Deltaforge 658 con Picos Móviles (Aplication of the Lubricant Deltaforge 658 with Moving Nozzles)*. Report # I 102-504/92, 1992. FUDETEC, Campana, Argentina.
5. Mendez, P.F. and Herrera, R., *Coeficientes de Fricción en el Conformado en Caliente (Friction Coefficients in Hot Metalworking)*. Report # I 1.72-538/92, 1992. FUDETEC, Campana, Argentina.
6. Mendez, P.F., Pochtaruk, M., and Herrera, R., *Análisis de Sensibilidad en la Aplicación de Lubricantes para Forja (Sensitivity Study of the Application of Forging Lubricants)*. Report # 1.72-414, 1992. FUDETEC, Campana, Argentina.
7. Carcagno, G., Daubian, P., Friederwitzer, P., Herrera, R., Johnson, D., Mendez, P.F., Morgenstern, V., Nicolosi, A., Pochtaruk, M., and Ugarriza, J., *Sistema Universal de Ensayo CTM (Universal Testing System CTM)*. Report # I-0.06-458/92, 1992. FUDETEC, Campana, Argentina.
8. Mendez, P.F., *Influencia de los Parámetros de Diseño de Un Regulador (Influence of Design Parameters of a Centrifugal Governor)* 1991. Universidad de Buenos Aires, Buenos Aires, Argentina.
9. Mendez, P.F., *Analisis de los Sistemas Dinámicos de Un Grado de Libertad que Incluyen Rozamiento (Analysis of Dynamic Systems of One Degree of Freedom that Include Friction)* 1989. Universidad de Buenos Aires, Buenos Aires, Argentina.
10. Mendez, P.F., *Verificaciones Experimentales de las Predicciones Teóricas Acerca del Comportamiento Estacionario y Transitorio de un Regulador Centrífugo (Experimental Verification of the Theoretical Predictions about the Steady-State and Transient Behavior of a Centrifugal Governor)* 1989. Universidad de Buenos Aires, Buenos Aires, Argentina.
11. Mendez, P.F., *Conceptos del Funcionamiento de los Reguladores Centrífugos de Velocidad (Fundamentals of Centrifugal Governors)* 1988. Universidad de Buenos Aires, Buenos Aires, Argentina.

Discussion of P. Mendez's work

1. Bhadeshia, H.K.D.H., *Problems in the Welding of Automotive Alloys.* Science And Technology Of Welding And Joining, 2015. **20** (6): p. 451-453.
2. Barasaba, N., *D.B. Robinson Distinguished Speaker Series Kicks-off with Dr. Latanision*. University of Alberta, Chemical and Materials Engineering website, Sep. 17, 2013. Edmonton, AB
3. Barasaba, N., *MSc student wins IIW Henry Granjon Prize for Joining and Fabrication Technology*. University of Alberta, Chemical and Materials Engineering website, May 6, 2013. Edmonton, AB
4. Flakstad, N., *Working to beat the best.* Oilsands Review, 2012 (March).
5. AITF. *AMFI Announcement.* in http://www.youtube.com/watch?v=4nWeAJ76rzI. Oct. 13, 2011. Edmonton, AB.
6. Cairney, R., *'Nimble' research could speed up advancements in welding*. University of Alberta, Chemical and Materials Engineering website, Dec. 15, 2011. Edmonton, AB
7. Cameron, K., *Welding lab sparks engineering students' interest*. University of Alberta, Chemical and Materials Engineering website, Feb. 1, 2011. Edmonton, AB
8. Murphy, B., *Joining hunt for a better weld*. University of Alberta website, Oct. 17, 2011. Edmonton, AB
9. Office of Government & Corporate Relations, *Welding centre seeks real-world solutions to industry challenges*. University of Alberta website, Nov. 5, 2011. Edmonton, AB
10. Sherk, L., *AMFI Program in Grande Prairie*. Peace Region Manufacturers Association (PERMA) Newsletter, November, 2011. Grande Prairie, AB
11. David, S.A., DebRoy, T., and Bhadeshia, H.K.D.H., *1000 gems: celebration of STWJ.* Science And Technology Of Welding And Joining, 2011. **16** (4): p. 285-287.
12. Bhadeshia, H.K.D.H., *Editorial.* Science And Technology Of Welding And Joining, 2010. **15** (8): p. 646-647.
13. Canadian Manufacturing staff, *The future is metal*. Canadian Manufacturing, Nov. 16, 2010.
14. Cairney, R., *Flashes of brilliance. New welding chair addresses and economic hot spot.* U of A Engineer, 2009 (27): p. 5.
15. Journal Business Staff, *Research chair aims to revolutionize welding*. The Edmonton Journal, May 13, 2009. Edmonton, AB
16. Kamath, C., *Editorial: Application-driven Data Analysis.* Statistical Analysis and Data Mining, 2008. **1** (5): p. 285.
17. Weilert, T., *Friday free pour at Hill Hall*. The Oredigger, January 21, 2008. Golden, CO
18. Arizmendi, R., *Un sensacional fin de curso*. Diario Vasco, June 24, 2007. Tolosa-Goierri
19. Arizmendi, R., *Del Goierri a Colorado*. Diario Vasco, May 10, 2006. Tolosa-Goierri
20. Arizmendi, R., *Un prometedor acuerdo de colaboración. La Universidad del Goierri estrecha lazos de cooperación con la de Colorado*. Diario Vasco, January 16, 2005. Tolosa-Goierri
21. Ehrenman, G., *The Past - And Future.* Mechanical Engineering, 2004. **126** (Design Supplement): p. 2.
22. Aminoff, J., *Order of Magnitude Modeling May Lead to Low Cost Welding.* Industry Collegium Report, 1999. **16** (1): p. 5.